Correlation & Association Visualization

Workshop 4 — Module 6

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Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

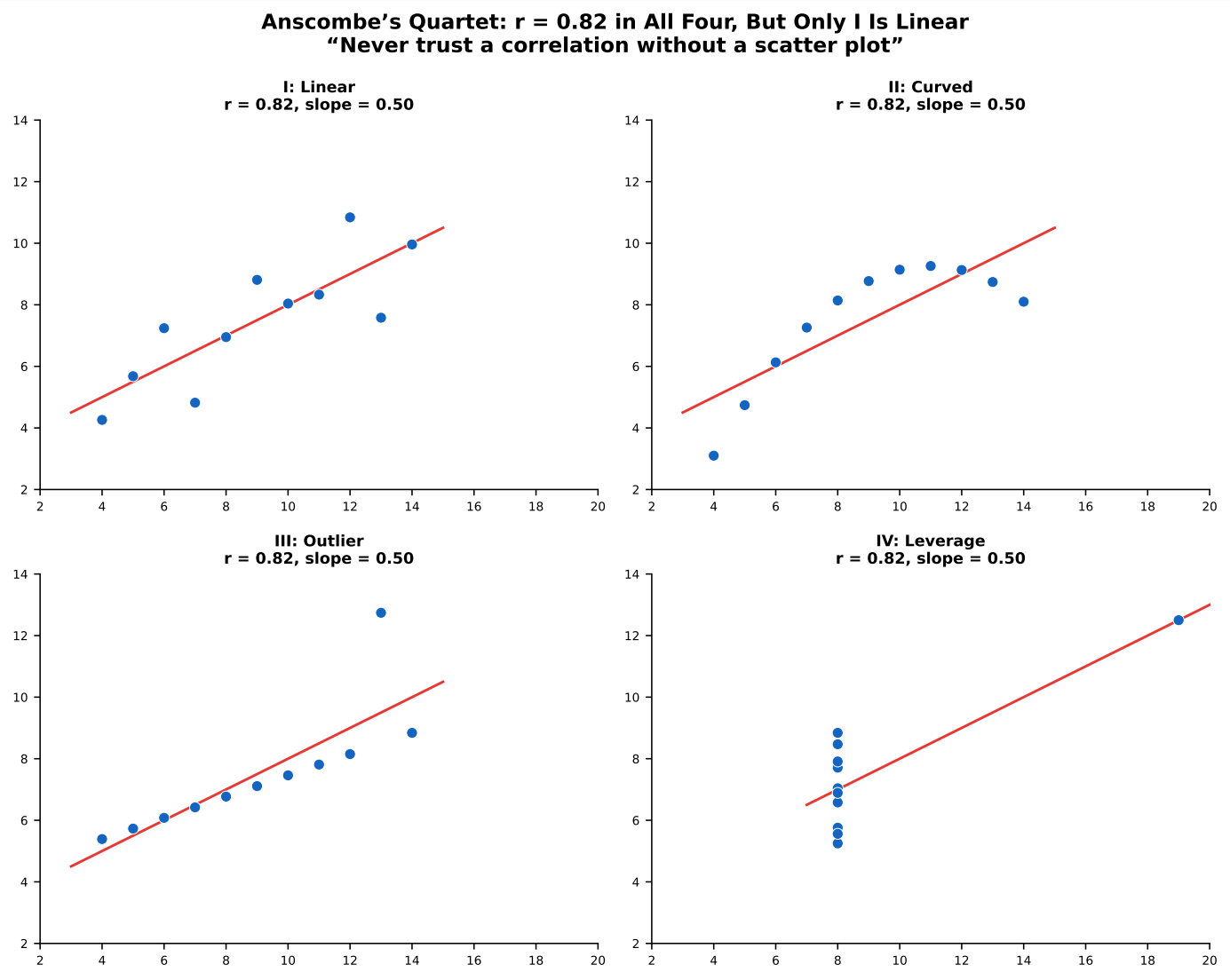
1. Anscombe's quartet: why scatter before correlate
2. Three correlation coefficients: Pearson, Spearman, Kendall
3. Correlation heatmaps: design principles
4. Pairs plots / scatterplot matrices
5. Categorical association: Cramér's V and mosaic plots
6. Correlation $\neq$ causation

Learning Outcome

You will always produce a scatter plot before computing a correlation coefficient, build publication-quality correlation heatmaps and pairs plots, and correctly visualize categorical association.

Anscombe's Quartet

Same r, Completely Different Stories



The Lesson

All four datasets have:

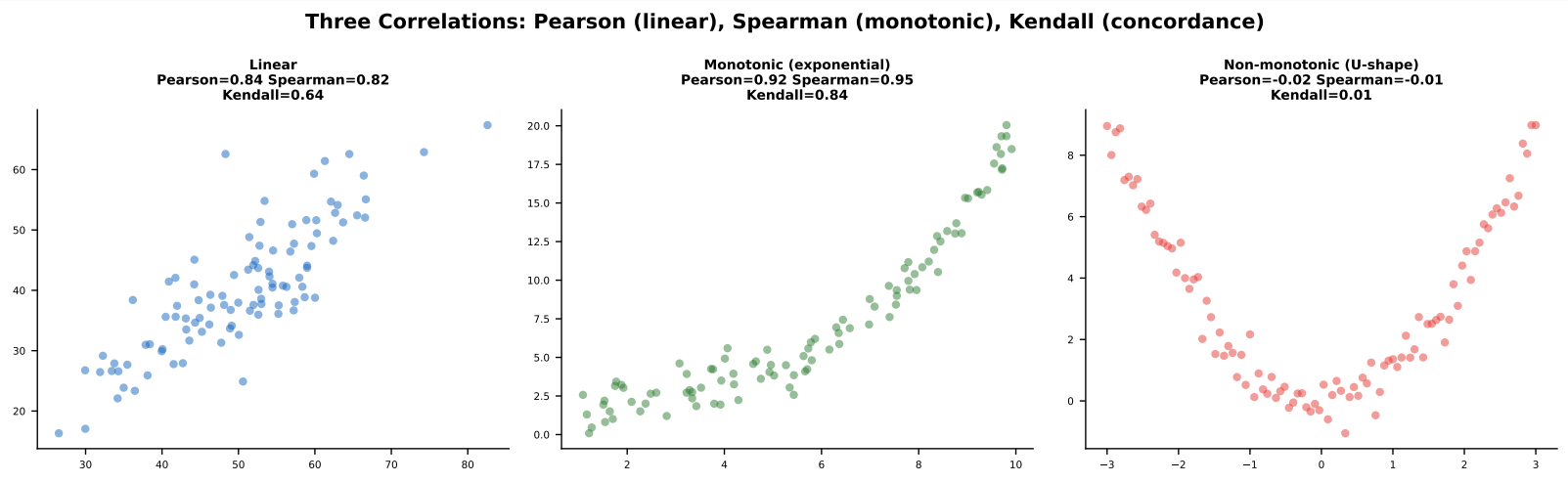
- $r = 0.82$, slope $= 0.50$, intercept $= 3.00$
- Same mean of x, same mean of y, same R^2

But only Dataset I is actually linear. II is curved (needs a quadratic); III has one outlier driving the fit; IV has one leverage point creating the illusion of correlation.

Pro-Tip
Rule: always scatter plot before computing any correlation. A single number (r) cannot capture curvature, outliers, clusters, or leverage. Anscombe (1973) designed these four datasets specifically to make this point.

Three Correlation Coefficients

Pearson, Spearman, Kendall



When to Use Which

Coefficient	Measures	Use When
Pearson r	Linear association	Both variables continuous, roughly
Spearman ρ	Monotonic association (rank-based)	Ordinal data, skewed, or nonlinear
Kendall τ	Concordance of pairs	Small n, many ties, or robustness

For the **U-shaped** relationship (right panel), all three return ≈ 0 — none captures non-monotonic association. Use **distance correlation** (dcor) or **mutual information** for arbitrary dependence.

Computing Correlations

R

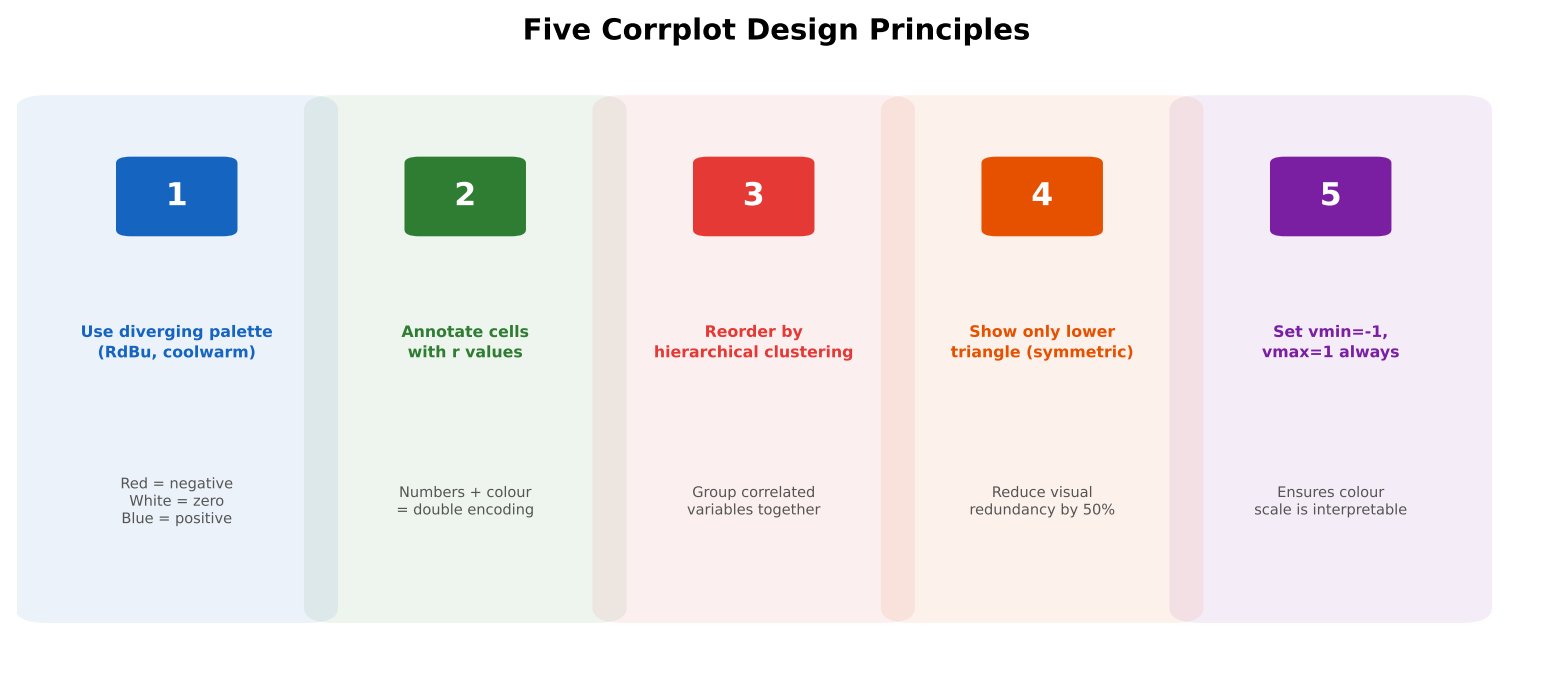
```r
# Full matrix
cor(df[, num_cols])                    #
     Pearson
cor(df[, num_cols],
    method = "spearman")               #
         Spearman

# Single pair with p-value
cor.test(df$x, df$y)
cor.test(df$x, df$y,
    method = "kendall")

# Robust: winsorised
library(WRS2)
wincor(df$x, df$y, tr = 0.1)

# Chi-squared for categorical
chisq.test(table(df$cat1,
                 df$cat2))
```

Python

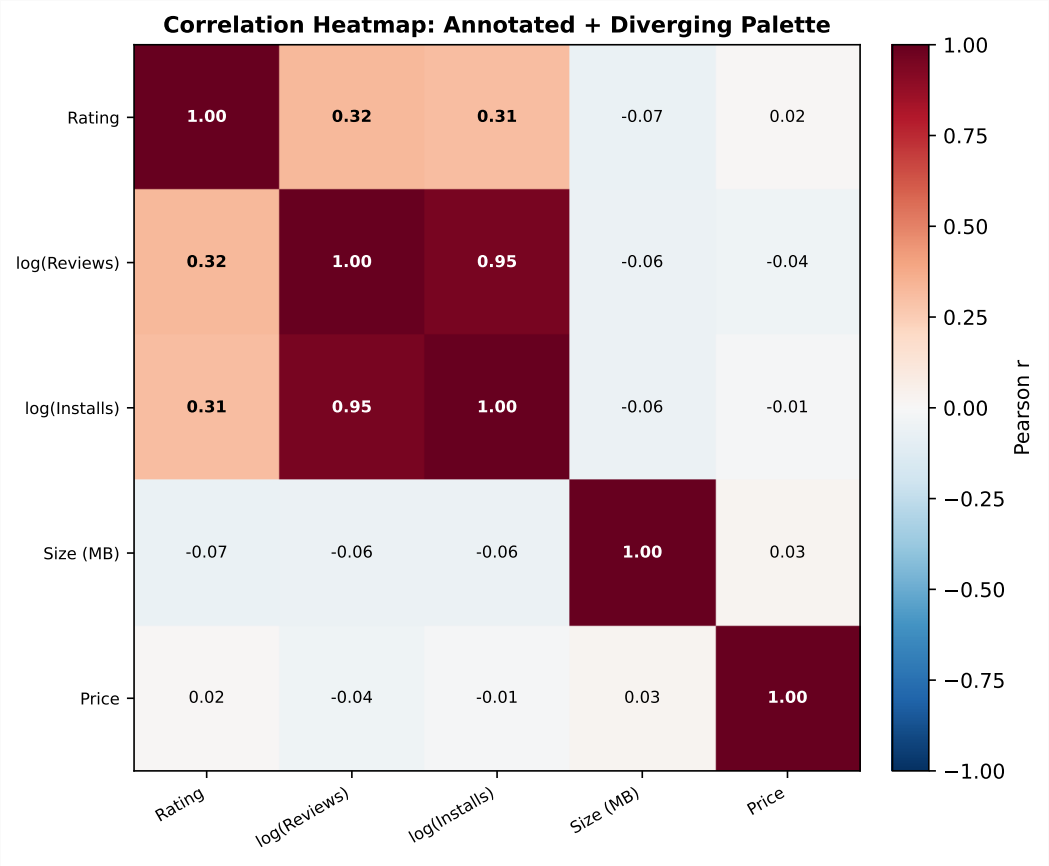
```python
# Full matrix
df[num_cols].corr()                    #
     Pearson
df[num_cols].corr(
    method="spearman")                 #
         Spearman

# Single pair with p-value
from scipy.stats import (
    pearsonr, spearmanr,
    kendalltau)
r, p = pearsonr(x, y)
rho, p = spearmanr(x, y)

# Chi-squared for categorical
from scipy.stats import (
    chi2_contingency)
ct = pd.crosstab(df.cat1, df.cat2
    )
chi2, p, dof, exp = (
    chi2_contingency(ct))
```

Correlation Heatmaps

Five Design Principles



Annotated Heatmap



Heatmap in Code

R (ggcorrplot or corrplot)

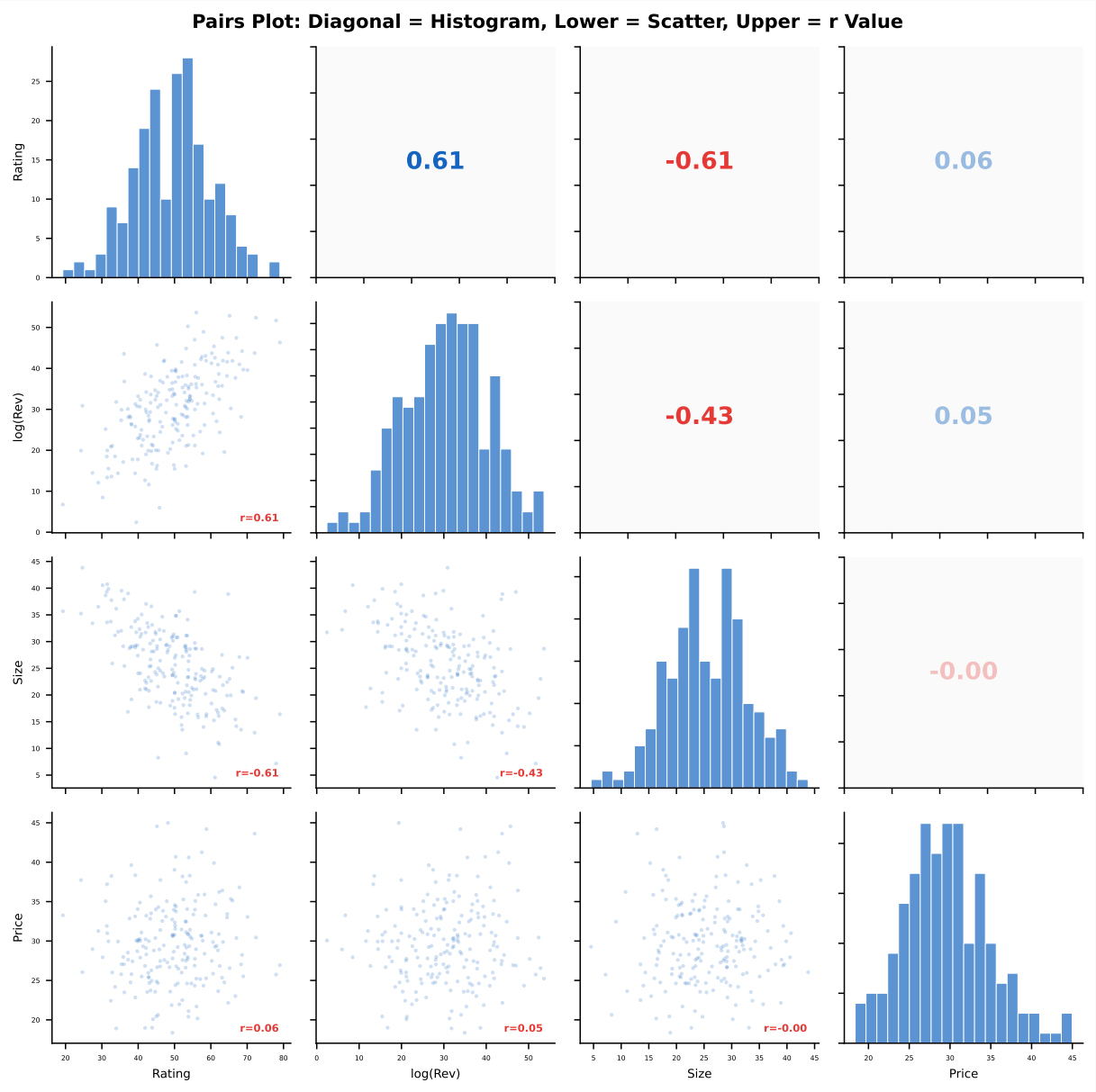
```r
library(ggcorrplot)

corr <- cor(df[, num_cols],
    use = "pairwise.complete")

ggcorrplot(corr,
  type = "lower",          #
      triangle
  lab = TRUE,              #
      annotate
  lab_size = 3,
  colors = c("#E53935",    # neg
    "white",               # zero
    "#1565C0"),            # pos
  hc.order = TRUE,         #
      cluster
  outline.color = "white")

# Alternative: corrplot package
library(corrplot)
corrplot(corr, method = "color",
  type = "lower", tl.col = "black
      ",
  addCoef.col = "black",
```

Python (seaborn)

```python
corr = df[num_cols].corr()

# Lower triangle mask
mask = np.triu(np.ones_like(
    corr, dtype=bool))

sns.heatmap(
    corr,
    mask=mask,                 #
        triangle
    annot=True,                #
        annotate
    fmt=".2f",
    cmap="RdBu_r",            #
        diverging
    vmin=-1, vmax=1,           #
        fixed scale
    center=0,
    square=True,
    linewidths=0.5,
    cbar_kws={"shrink": 0.8,
        "label": "Pearson r"})
```

Pairs Plots

The Scatterplot Matrix



Pairs Plot in Code

R (GGally)

```r
library(GGally)

ggpairs(df,
  columns = c("Rating",
    "log_reviews", "Size"),
  mapping = aes(color = Type,
    alpha = 0.3),
  lower = list(continuous =
    wrap("points", size = 0.5)),
  diag = list(continuous =
    wrap("densityDiag")),
  upper = list(continuous =
    wrap("cor", size = 3))) +
  theme_minimal(base_size = 8)
```

Python (seaborn)

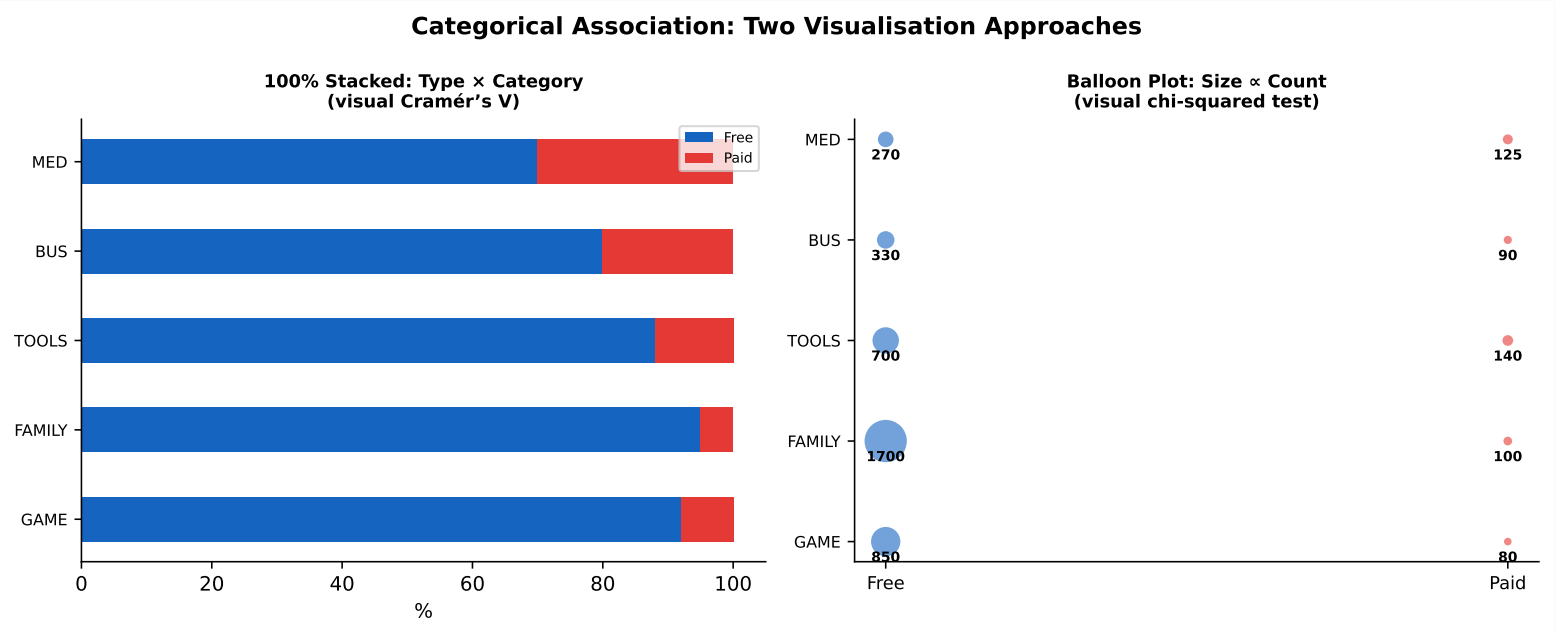
```python
sns.pairplot(
    df[["Rating", "log_reviews",
        "Size", "Type"]],
    hue="Type",
    palette={"Free": "#1565C0",
             "Paid": "#E53935"},
    diag_kind="kde",
    plot_kws={"s": 10,
              "alpha": 0.3},
    height=2)
```

Pro-Tip

Pairs plots scale poorly beyond ~8 variables (64 panels). For high-dimensional data, use a **correlation heatmap** first to identify the strongest pairs, then drill down with individual scatter plots or pairs on a subset.

Categorical Association

Visualizing Categorical × Categorical



Data Insight

For two categorical variables, the visual equivalent of a chi-squared test is a **100% stacked bar** (proportions) or a **balloon plot** (counts, size $\propto$ count). If all bars look identical, the variables are independent. Differences in bar composition $\Rightarrow$ association. Quantify with **Cramér's V** $\in [0, 1]$.

Correlation $\neq$ Causation

The Most Important Slide in This Module

Correlation ≠ Causation: Four Classic Spurious Examples

Variable A	Variable B	Explanation	Type
Ice cream sales ↑	Drowning deaths ↑	Summer heat (confound)	
Shoe size ↑	Reading ability ↑	Age (confound)	
Nicolas Cage films ↑	Pool drownings ↑	Pure coincidence (spurious)	
Storks counted ↑	Birth rate ↑	Rurality (ecological fallacy)	

Pro-Tip

Three sources of spurious correlation: **confounders** (a third variable causes both), **reverse causation** (Y causes X, not X causes Y), and **coincidence** (random co-trend in finite samples). Correlation establishes association; only controlled experiments or careful causal inference (DAGs, instrumental variables) establish causation.

Synthesis

Key Takeaways

1. **Always scatter first**. Anscombe's quartet proves that r alone is meaningless without a visual check.
2. **Pearson** = linear; **Spearman** = monotonic; **Kendall** = concordance. None captures non-monotonic dependence — use distance correlation for that.
3. Heatmap design: **diverging palette, annotate cells, cluster rows/columns, lower triangle only, fixed** $[-1, 1]$ **colour scale**.
4. **Pairs plots** combine scatter + density + r in one figure. Limit to ≤ 8 variables.
5. For categorical association: **100% stacked bar** or **balloon plot** + Cramér's V.
6. **Correlation** $\neq$ **causation**. Always consider confounders, reverse causation, and coincidence.

Essential Readings

- Anscombe, F. J. (1973). "Graphs in Statistical Analysis." *The American Statistician*, 27(1), 17–21.
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 12 (Visualizing Associations).
- Friendly, M. (2002). "Corrgrams." *The American Statistician*, 56(4), 316–324.
- Pearl, J. (2009). *Causality*, 2nd ed. Cambridge University Press. (Correlation vs causation.)

Next Module: EDA Case Study — Google Play Store (W04-M07)